

Experiment No 11

AIM: Study of Half wave rectifiers

APPARATUS: Rectifier Kit, Digital Multimeter, Connecting wires.

THEORY:

In Half Wave Rectification, When AC supply is applied at the input, only Positive Half Cycle appears across the load whereas, the negative Half Cycle is suppressed. How this can be explained as follows:

During positive half-cycle of the input voltage, the diode D1 is in forward bias and conducts through the load resistor R_L . Hence the current produces an output voltage across the load resistor R_L , which has the same shape as the +ve half cycle of the input voltage.

During the negative half-cycle of the input voltage, the diode is reverse biased and there is no current through the circuit. i.e., the voltage across R_L is zero. The net result is that only the +ve half cycle of the input voltage appears across the load. The average value of the half wave rectified o/p voltage is the value measured on dc voltmeter.

For practical circuits, transformer coupling is usually provided for two reasons.

1. The voltage can be stepped-up or stepped-down, as needed.
2. The ac source is electrically isolated from the rectifier. Thus preventing shock hazards in the secondary circuit.

The efficiency of the Half Wave Rectifier is 40.6%

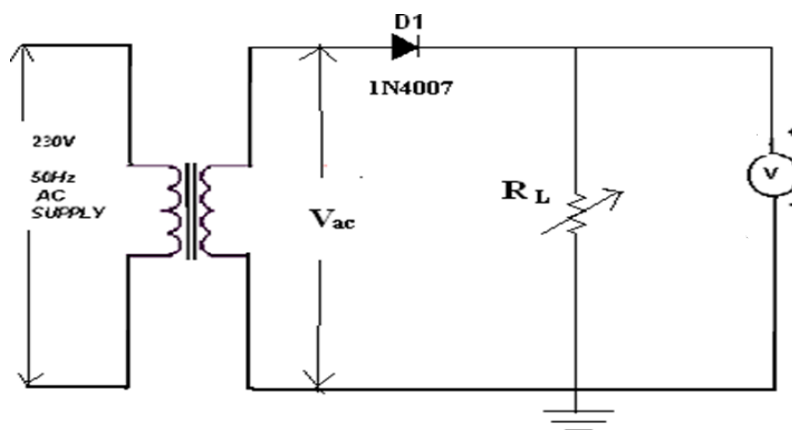
Theoretical calculations for Ripple factor: Without Filter:

$$V_{rms} = V_m/2 \quad V_m = 2V_{rms} \quad V_{dc} = V_m/\pi$$

$$\text{Ripple factor } r = \sqrt{(V_{rms}/V_{dc})^2 - 1} = 1.$$

CIRCUIT DIAGRAM:

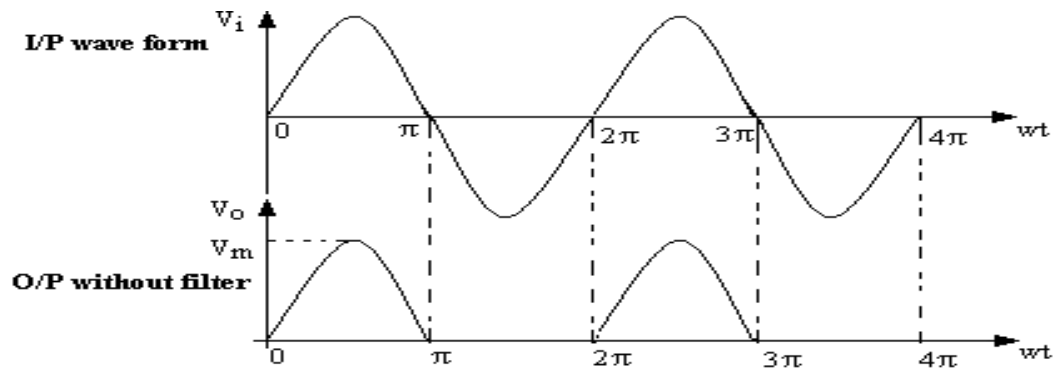
Half wave Rectifier



Observation Table

Sr. No	Load Resister (RL)	vrms Value of AC signal	DC Voltage (Vdc)	Ripple Factor (γ)
1	47 Ω			
2	100 Ω			

MODEL WAVEFORMS:



PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.
3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output of the rectifier.
4. Find the theoretical of dc voltage by using the formula,
$$V_{dc} = V_m / \pi$$

Where, $V_m = 2V_{rms}$, (V_{rms} = output ac voltage.)
5. The Ripple factor is calculated by using the formula
$$r = \text{ac output voltage} / \text{dc output voltage}.$$

RESULT: Hence we are study of half wave rectifier and find its ripple factor.

PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified
2. The polarities of all the diodes should be carefully identified.
3. While determining the % regulation, first Full load should be applied and then it should be decremented in steps.

VIVA QUESTIONS:

1. What is the PIV of Half wave rectifier?
2. What is the efficiency of half wave rectifier?
3. What is the rectifier?
4. What is the difference between the half wave rectifier and full wave Rectifier?
5. What is the o/p frequency of Bridge Rectifier?
6. What are the ripples?
7. What is the function of the filters?